

UNIVERSITÀ CA' FOSCARI

Master's Degree Programme (DM270)

Statistical Models & Methods for Finance

Homework: Final Exam

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1 Introduction

The document aims to analyze the daily difference in Tesla Inc. stock prices with the purpose of outlining the exact thought process of the time-series analysis and model evaluation. The document concludes by synthesizing the evidence presented.

2 Analysis of TSLA Daily Difference Time-Series

The time series is downloaded from Yahoo Finance and represents the daily difference of Tesla Inc. (ticker TSLA) over the period from 2024-10-01 to 2026-01-12. Importantly, the daily difference is defined as the difference between the maximum and minimum values within a trading day and may thus be interpreted as a measure of *daily uncertainty*.

The raw data, together with its autocorrelation and partial autocorrelation functions (ACF and PACF), is shown in Figure 1. Preliminarily, the time series of daily differences appears to fluctuate non-randomly around a constant mean, roughly between 10 and 20. Furthermore, when the daily difference is low, it seems to persist in that range, and vice versa. Notwithstanding these initial observations, a more formal analysis is needed for confirmation. From the ACF plot, the pattern seems to decay somewhat gradually, with a few incremental spikes exceeding the significance bounds at lag 5 and lags 9–10. The PACF shows two spikes above the confidence bounds, although the second lies close to the bounds and may reflect random variation. Combined with the apparent decay in the ACF, this pattern may indicate an AR(2) process, or alternatively an AR(1) process.

Since the stationarity condition of any ARMA(p, q) process, including AR(p) processes, depends on the roots of the autoregressive characteristic polynomial, an Augmented Dickey–Fuller (ADF) test for a unit root is performed on the raw time-series. The null hypothesis is the presence of a unit root (non-stationarity), whereas the alternative hypothesis is the absence of a unit root (stationarity). The test is conducted without trend and without drift, and the results are reported in Listing 1. The observed test statistic is -2.7103 and is more negative than the corresponding critical values at the 1%, 5%, and 10% significance levels (-2.58, -1.95, and -1.62, respectively). Consequently, the null hypothesis is rejected at all conventional significance levels, providing empirical evidence against the null; that is, the time-series appear stationary. For this reason, no further transformation is applied. [1]

```
# Augmented Dickey--Fuller test (no constant, no trend)
> summary(adf_raw)

Value of test-statistic: -2.7103
Critical values: 1% = -2.58, 5% = -1.95, 10% = -1.62

Coefficients:
z.lag.1      = -0.06452  (p = 0.00709)
z.diff.lag   = -0.36658  (p < 1e-10)
```

[1] When using adjusted prices, one would typically analyze log-returns, which correspond to continuously compounded returns and are standard in time-series modeling. However, since this assignment explicitly focuses on daily differences, a logarithmic transformation has been deemed unnecessary.

Listing 1: ADF Test Output for Daily Difference

3 Model Fitting & Diagnostics

The time series of daily differences in Tesla Inc. stock prices represents only a partial realization of a single trajectory from an underlying stochastic process. Consequently, any proposed model may, at best, only represent a reasonable approximation rather than a definitive conclusion of true stochastic process. For this reason, it is essential to evaluate several candidate models alongside their associated diagnostics.

This assignment considers three model specifications: an AR(1), an AR(2), and an ARMA(1,1) model. The AR(1) and AR(2) models are motivated by the preliminary inspection of the raw time-series, while the ARMA(1,1) model is selected using the `auto.arima()` function in `R` based on the Bayesian Information Criterion (BIC). The use of BIC reflects an explicit modeling choice and is a personal preference, as it tends to favor more parsimonious models, although the Akaike Information Criterion (AIC) could alternatively have been utilized.

For a linear time-series model to be appropriate, its errors must behave like white noise: zero mean, constant variance, and no autocorrelation at non-zero lags. Since the true error process is unobservable, residuals are examined instead. Diagnostics for all three models are presented in Figures 2–4, which include standardized residual plots, residual ACFs, and Ljung–Box test p-values. The Ljung–Box test is a simultaneity test with the null hypothesis that all autocorrelations up to lag h are jointly equal to zero, and the alternative that at least one autocorrelation differs from zero. Importantly, this is not equivalent to stating that all autocorrelations are different from zero. Figure 2, corresponding to the AR(1) model, shows that the Ljung–Box p-values fall below the 5% significance level from lag 2 onward, leading to rejection of the null hypothesis and providing empirical evidence of residual autocorrelation. The diagnostics of the AR(1) model are therefore deemed unsatisfactory. In contrast, Figures 3 and 4, corresponding to the AR(2) and ARMA(1,1) models, display p-values that consistently exceed the 5% significance level, generally remaining above 0.4. Thus, the residuals appear consistent with the null hypothesis of no autocorrelation, and the null cannot be rejected. Consequently, the diagnostics of both the AR(2) and ARMA(1,1) models appear satisfactory. The subsequent analysis therefore focuses on these two specifications.

Estimation results for the AR(2) and ARMA(1,1) models are reported in Listings 2 and 3. Both AIC and BIC are marginally lower for the AR(2) model relative to the ARMA(1,1) model (AIC: $2148.72 < 2148.85$; BIC: $2163.81 < 2163.94$), although the differences are small. This suggests that the AR(2) model offers a slightly better trade-off between fit and complexity. This finding is noteworthy given that the ARMA(1,1) model was selected by `auto.arima()` based on BIC, underscoring that automated procedures should be viewed as complements rather than definitive selection tools. Across both models, the estimated mean value is nearly identical (15.9292 for AR(2) and 15.9185 for ARMA(1,1)), consistent with the statements on the preliminary analysis on the raw time series. Regarding dynamics, the ARMA(1,1) model features a larger first-order autoregressive coefficient, likely compensating for the absence of a second autoregressive

term, whereas the moving-average coefficient is negative, indicating short-run adjustment. Lastly, to assess whether either of the two models includes unnecessary lags, hypothesis tests were conducted on the lag coefficients in both models to evaluate whether they are statistically different from zero. This inference is valid because the model diagnostics are satisfactory, allowing reliance on the asymptotic distribution of the estimators, assuming that the sample size (321 observations) is sufficiently large. For both models, all coefficients appear to be statistically different from zero at the 5% significance level, and therefore neither model seems to contain redundant coefficients. [2] Overall, both models appear robust, and the analysis therefore proceeds with both specifications when evaluating predictive performance.

```
Series: TSLA_train
ARIMA(2,0,0) with non-zero mean

Coefficients:
         ar1      ar2      mean
      0.3803   0.1787   15.9292
s.e.  0.0548   0.0548    0.8546

sigma^2 = 46.50
log likelihood = -1070.36
AIC = 2148.72   AICc = 2148.85   BIC = 2163.81
```

Listing 2: AR(2) Model Output

```
Series: TSLA_train
ARIMA(1,0,1) with non-zero mean

Coefficients:
         ar1      ma1      mean
      0.7451  -0.3715   15.9185
s.e.  0.0775   0.1113    0.9278

sigma^2 = 46.52
log likelihood = -1070.43
AIC = 2148.85   AICc = 2148.98   BIC = 2163.94
```

Listing 3: ARMA(1,1) Model Output

4 Predictive Performance

The predictive performance is evaluated out-of-sample using a test sample from 2026-01-13 to 2026-01-22. The test set contains six observations rather than nine, as markets are closed on weekends. The plotted 6-step-ahead forecasts with 95% prediction intervals are

[2] Under asymptotic normality of the MLE, we test $H_0 : \theta = 0$ using $z = \hat{\theta}/\,\mathrm{SE}(\hat{\theta})$ with a two-sided 5% critical value of 1.96. Observed z-statistics for dynamic coefficients: *AR(2)*: $(z_{\phi_1}, z_{\phi_2}) = (6.94, 3.26)$; *ARMA(1,1)*: $(z_\phi, z_\theta) = (9.61, -3.34)$. Since $|z_{obs}| > 1.96$, the null is rejected for all dynamic coefficients in both models. See code for calculations.

shown in Figures 5 and 6, and the corresponding forecast accuracy measures are reported in Listings 4 and 5. The figures indicate that the point forecasts and prediction intervals are broadly similar across the two models, making visual comparison inconclusive. Moreover, the prediction intervals rely on the restrictive assumption of Gaussian White Noise (GWN). For these reasons, greater emphasis is placed on the forecast accuracy measures.

Turning to the forecast accuracies, I focus primarily on the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), as these measures are not subject to offsetting effects from positive and negative forecast errors. One could also consider the Mean Absolute Percentage Error (MAPE). However, I avoid this metric since it is expressed in relative terms, whereas my preference is to compare models in absolute terms. From the results, it becomes clear that the ARMA(1,1) model performs slightly better than the AR(2) model in terms of both MAE ($5.378133 < 5.515804$) and RMSE ($5.890762 < 6.064780$). Nevertheless, these differences are marginal. Specifically, the differences amount to 0.137671 USD for MAE and 0.174018 USD for RMSE, both of which appear minor when compared to the summary statistics of the training and test data, which indicate an average daily difference of 15.93 USD and a range from 2.96 to 61.53 USD. Hence, both models appear to perform very similarly in terms of forecast accuracy.

```
> accuracy(predfit.AR2$mean, TSLA_test)

              ME       RMSE       MAE        MPE       MAPE
Test set   -4.652625   6.064780   5.515804   -54.74045   59.38619
```

Listing 4: Forecast Accuracy for AR(2) Model (Test Set)

```
> accuracy(predfit.bic$mean, TSLA_test)

              ME       RMSE       MAE        MPE       MAPE
Test set   -4.506169   5.890762   5.378133   -53.06078   57.7538
```

Listing 5: Forecast Accuracy for ARMA(1,1) Model (Test Set)

5 Assessment of Volatility Clustering

Having evaluated the raw time series, along with the model diagnostics and predictive performance of the AR(2) and ARMA(1,1) models, it becomes relevant to consider the potential existence of volatility clustering.

In the previous sections, the diagnostics for the AR(2) and ARMA(1,1) models were deemed satisfactory, in the sense that the residuals behave approximately as white noise. This justifies testing for volatility clustering by examining the squared residuals. While the residuals themselves are uncorrelated, ARCH and GARCH models allow for time-varying conditional variance, implying autocorrelation in squared residuals. In ARCH models, the conditional variance depends on lagged squared residuals, whereas in GARCH models it depends on both lagged squared residuals and lagged conditional variances.

Figures 7–8 present the diagnostics for volatility clustering. These plots include the time series of squared residuals, the ACF and PACF of the squared residuals, and the corresponding p-values from the McLeod–Li test. The McLeod–Li test is a test for autocorrelation in squared residuals up to lag h and serves the purpose of detecting potential volatility clustering. The null hypothesis is that all autocorrelations of the squared residuals are jointly equal to zero (no volatility clustering), while the alternative hypothesis is that at least one autocorrelation of the squared residuals is significantly different from zero (volatility clustering).

From both figures, it becomes evident from the ACF and PACF plots that none of the lags exhibit spikes outside the confidence bounds. Moreover, the McLeod–Li test p-values for both models remain above the 5% significance level across all lags. As a result, we fail to reject the null hypothesis, and therefore conclude that the squared residuals are consistent with the null hypothesis of no autocorrelation in squared residuals. Consequently, there is no empirical evidence of volatility clustering in the data for either model. Since neither model exhibits empirical signs of volatility clustering, no further analysis is pursued in this direction.

6 Discussion & Recommendation

This analysis attempted to analyze the daily price difference of Tesla Inc. stock over the period from 2024-10-01 to 2026-01-12. This measure has a clear economic interpretation, as it quantifies the *"daily uncertainty"* by capturing the difference between the daily maximum and minimum prices. Furthermore, the ADF unit root test provided empirical evidence against the null hypothesis, suggesting that the time-series is stationary. Consequently, the analysis proceeded without applying further transformations.

Based on the analysis, two final models were considered: an AR(2) model motivated by the ACF and PACF of the raw data, and an ARMA(1,1) model suggested by the `auto.arima()` function in `R` using BIC. Both models exhibited satisfactory diagnostic properties, with only minor differences in BIC and AIC values. Somewhat surprisingly, the AR(2) model achieved lower BIC and AIC values than the ARMA(1,1), despite the latter being selected by `auto.arima()`, underscoring that `auto.arima()` should be viewed as a complementary rather than definitive selection tool. Forecast performance differed only marginally, with the ARMA(1,1) model performing slightly better on the six out-of-sample test observations. Finally, neither model showed evidence of autocorrelation in squared residuals, indicating no volatility clustering. Overall, the differences between the models seem small, and the final choice therefore depends largely on practical considerations and personal preference.

On a personal note, the AR(2) model is preferred for several reasons. First, it is directly motivated by the ACF and PACF of the time series, making the rationale for model selection transparent. Second, the AR(2) model is arguably more interpretable than the ARMA(1,1) model, as it depends solely on its own lagged values rather than past errors. Lastly, the out-of-sample evaluation is based on only six test observations, and the differences in forecast accuracy are slight. These differences are therefore likely driven by random variation rather than systematic differences in predictive performance. Consequently, greater emphasis is placed on the in-sample information criteria, AIC and BIC. Taken together, these considerations motivate the final model choice.

A Appendix

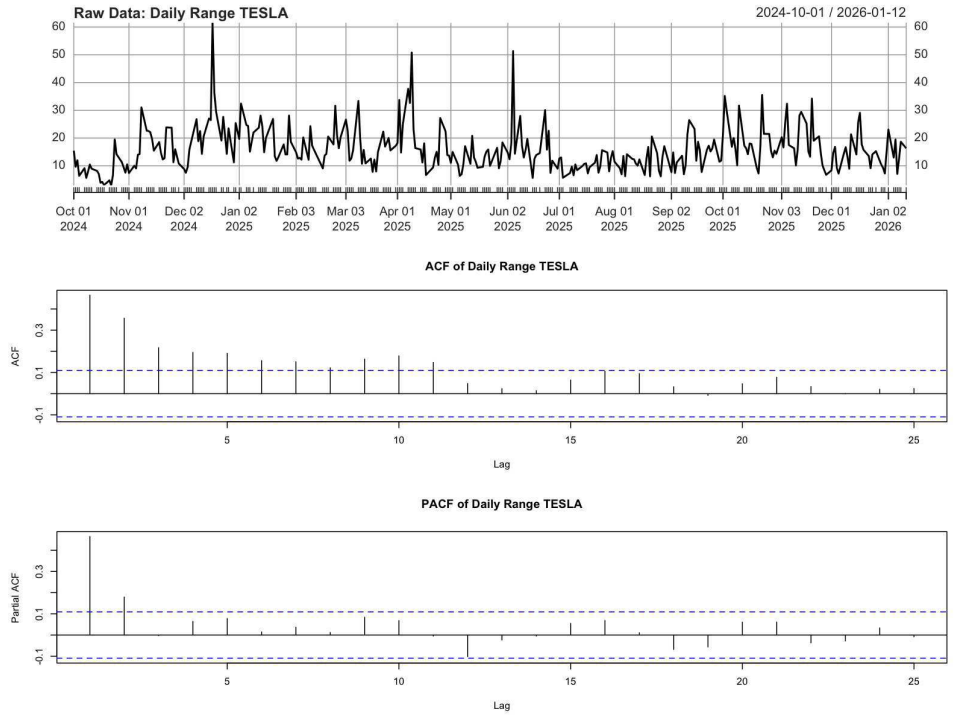


Figure 1: Time-Series Plot of Daily Range for Tesla Inc.

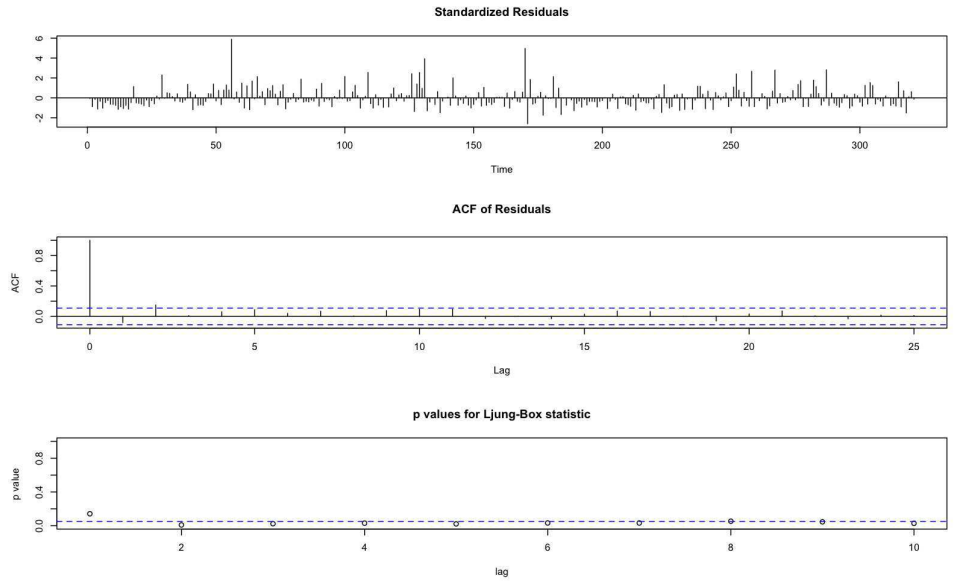


Figure 2: Diagnostics of Autoregressive Model of Order 1, AR(1)

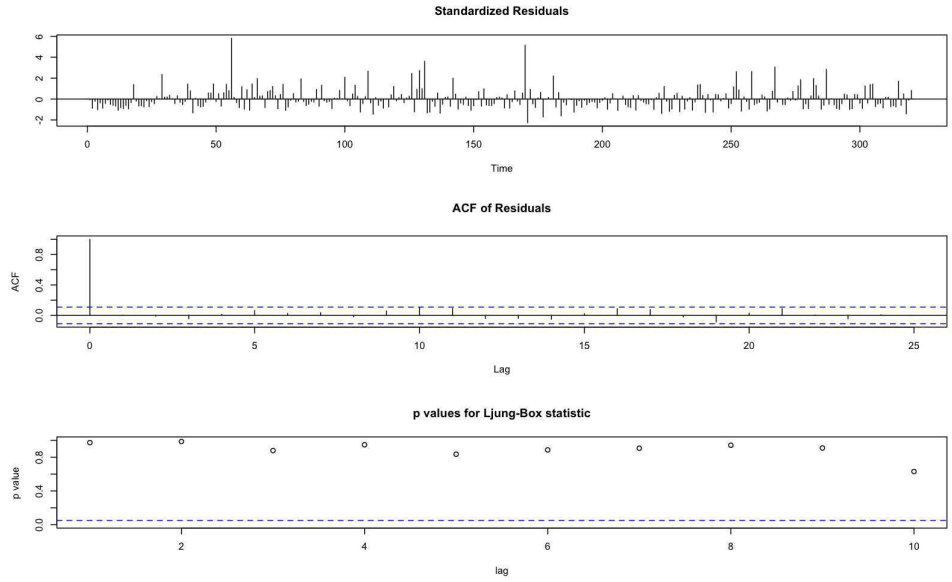


Figure 3: Diagnostics of Autoregressive Model of Order 2, AR(2)

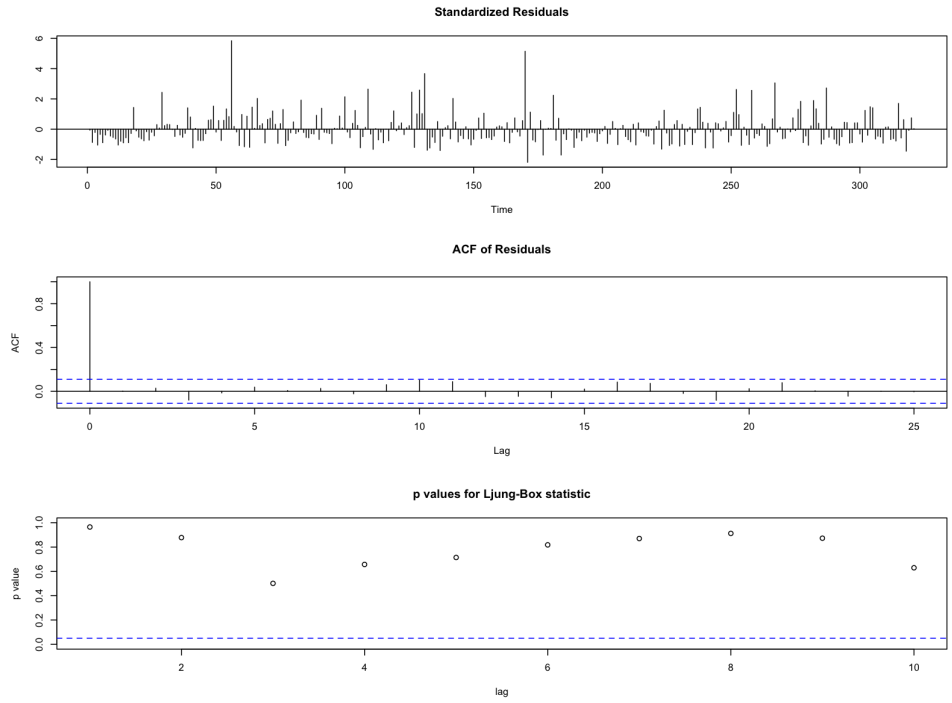


Figure 4: Diagnostics of ARMA(1,1) Model, Selected by BIC

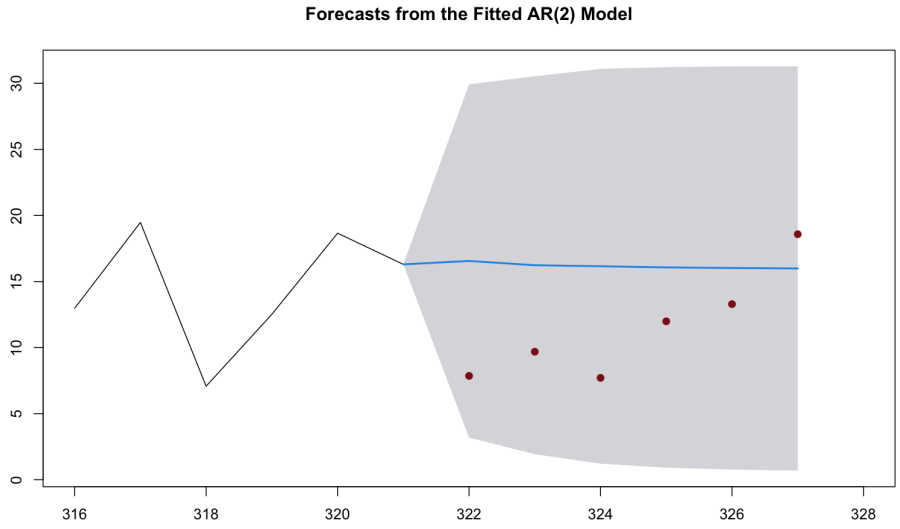


Figure 5: Predictive Performance of AR(2) Model

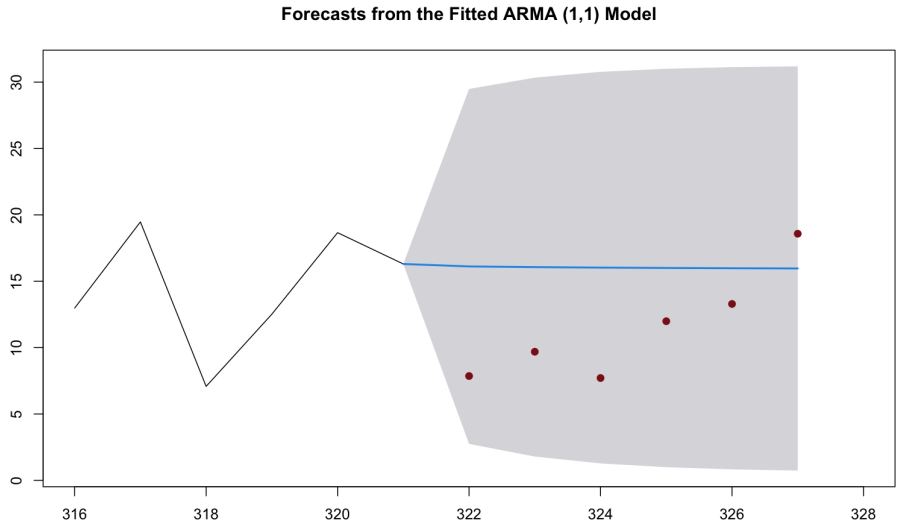


Figure 6: Predictive Performance of ARMA(1,1) Model

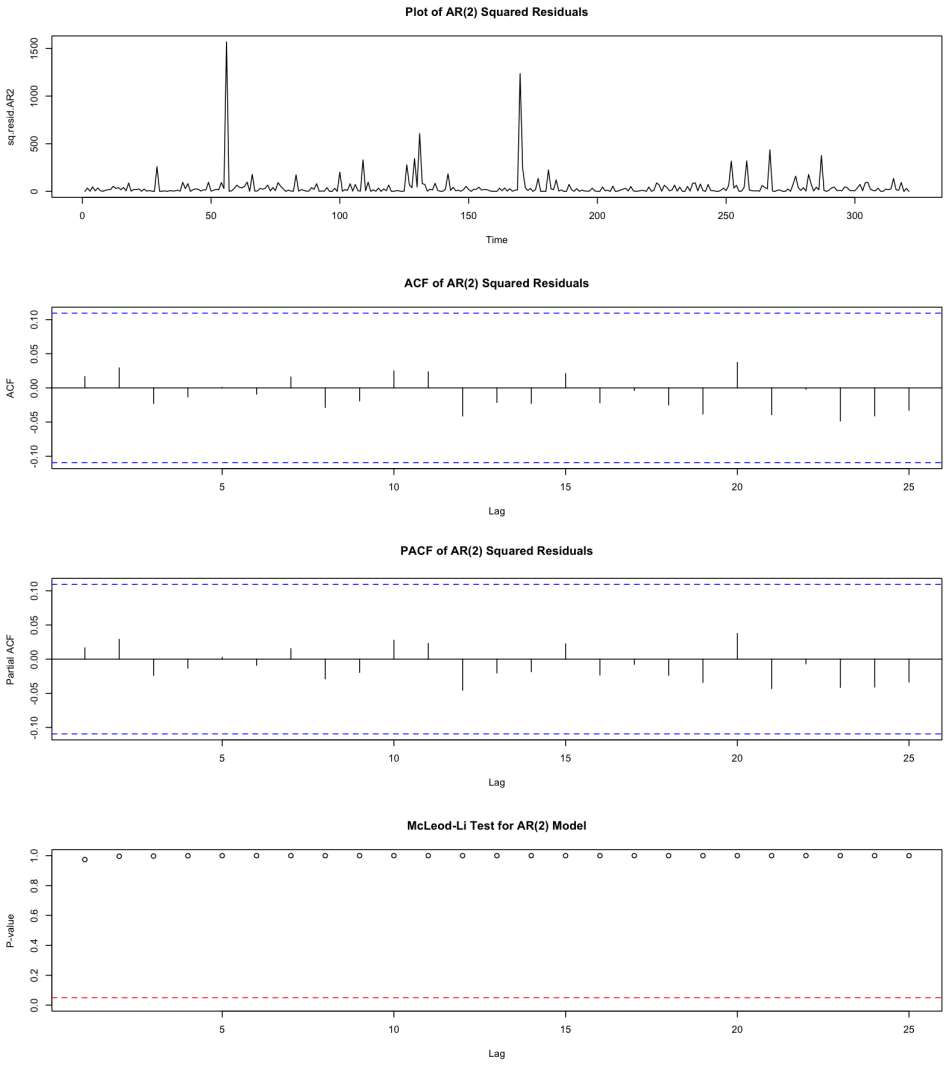


Figure 7: Evaluation of Volatility Clustering for AR(2) Model

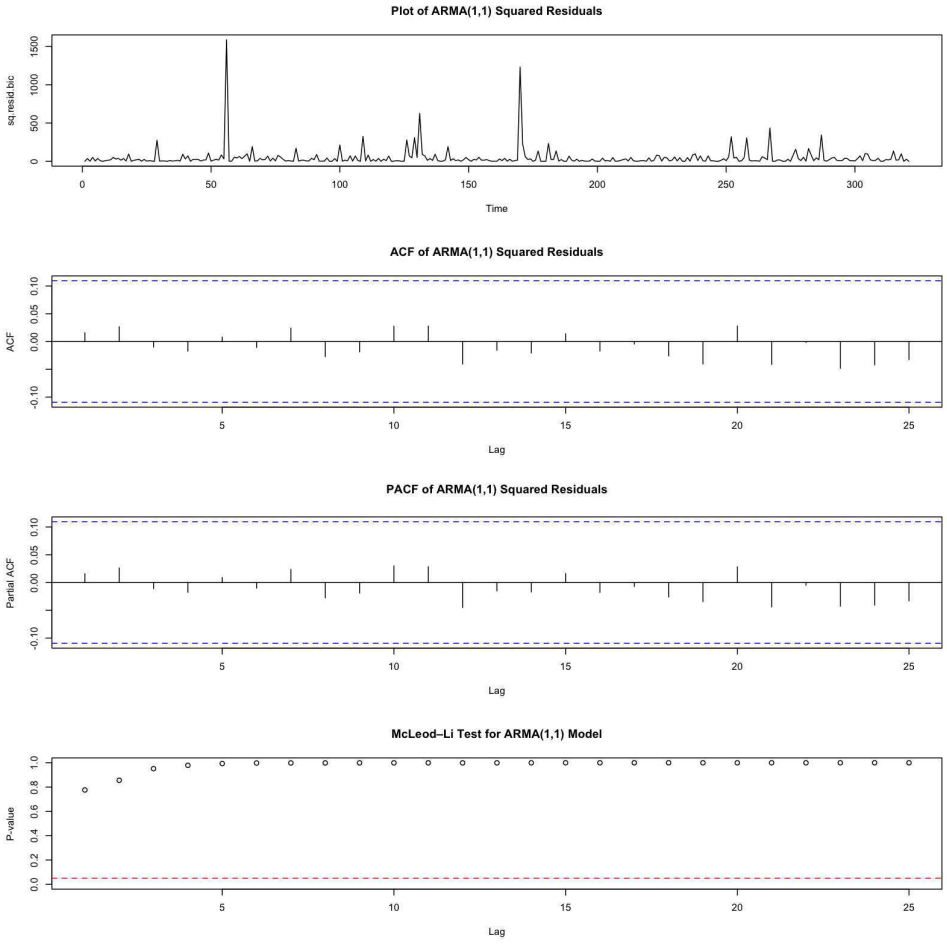


Figure 8: Evaluation of Volatility Clustering for ARMA(1,1) Model